Konstantin Fedorov

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Research Interests

NLP for Software Engineering: LLM-based agents, repository-level code generation, structure-aware retrieval, executable evaluation. **General NLP:** structured prediction, discrete diffusion models, information extraction.

Education

- **ITMO University, AI Talent Hub** *2025 – 2027 expected*
  M.Sc. in Artificial Intelligence Saint Petersburg, Russia
  - Current GPA: 5.0/5.0 (first semester).
- **Innopolis University** *2021 – 2025*
  B.Sc. in Computer Science, Data Science Specialization Innopolis, Russia
  - GPA: 4.69/5.00. Bachelor thesis: *Graph Retrieval Augmented Code Generation*; grade: A.
  - Thesis supervisor: Dr. Vladimir Ivanov.

Publications

[1] **Konstantin Fedorov**, Valentin Malykh. (2026). **DiffIE: Diffusion-based Open Information Extraction**. *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*. Oral presentation

[2] **Konstantin Fedorov**, Boris Zarubin, Vladimir Ivanov. (2025). **Graph Retrieval Augmented Code Generation**. *MAS-GAIN Workshop, ASEW 2025*. DOI: 10.1109/ASEW67777.2025.00060. Oral presentation.

Research Experience

- **ITMO University** *2025 – Present*
  Graduate Student Researcher (advisor: Dr. Valentin Malykh) Saint Petersburg, Russia
  - Proposed DiffIE, reformulating Open Information Extraction as conditional discrete diffusion over per-token role tags, which makes the extraction budget a tunable inference-time choice.
  - Achieved new state of the art on CaRB (1-1) in both F1 and AUC and the best average performance across four OpenIE benchmarks; first-author paper under review at EMNLP 2026 (ACL Rolling Review) [U.1].
- **Institute for System Dynamics and Control Theory, SB RAS** *Oct 2025 – Present*
  Junior Researcher (full-time) Irkutsk / Remote, Russia
  - Designing an LLM-based agent for scientific GIS workflows that translates natural-language requests into SQL queries over PostgreSQL databases and produces maps, tables, and analytical outputs.
  - Prototyping constrained code generation for geospatial processing: the model writes Python functions against predefined FastAPI service signatures executed in sandboxed Docker environments.
  - Contributing to Sentinel-2 temporal modeling with ConvLSTM-style approaches for remote-sensing analysis.
- **NLP for Code Repositories Laboratory, Innopolis University** *Apr 2024 – Jul 2024*
  ML Research Intern Innopolis, Russia
  - Built a repository-aware RAG pipeline for automatic README generation, including AST-based extraction of cross-file dependencies and external API usage traces.
  - Early exploration that informed the structure-aware retrieval direction later developed in GRACG, under the supervision of Dr. Vladimir Ivanov.

Selected Research and Engineering Projects

- **GRACG: Graph Retrieval Augmented Code Generation** *2024 – 2025*
  Tools: PyTorch Geometric, GraphSAGE, heterogeneous graphs, AST parsing, NetworkX, UniXcoder, Docker
  - Designed a graph-augmented retrieval system for repository-level code generation using AST-derived heterogeneous repository graphs and GraphSAGE-based representation learning.
  - Evaluated structure-aware retrieval against dense UniXcoder retrieval on RepoBench and DevEval-derived tasks, achieving 2–3 point improvements on retrieval metrics over strong dense baselines.
  - Transformed DevEval into a Dockerized EvoCodeBench-style benchmark to support executable evaluation of generated repository-level code.
  - Published at MAS-GAIN Workshop, ASEW 2025 (oral). Extends bachelor's thesis (grade A).
- **RAGCore: Repository-Level QA Framework** *2025*
  Tools: Python, Qdrant, Docker, AST chunking, reranking, LLM-as-judge evaluation

- Built an open-source framework for codebase question answering over GitHub repositories with AST-based chunking, Qdrant-backed retrieval, reranking, context expansion, and Docker-based deployment.
- Added agentic orchestration and LLM-as-judge evaluation for iterative codebase reasoning and retrieval quality assessment.
- Integrated SWE-QA as the benchmark for repository-level QA evaluation, with a merged pull request to the SWE-QA repository.

- **LLM-Based GIS Workflow Agent and Service Composition Prototype** *2025 – Present*
  Tools: LLM agents, PostgreSQL, FastAPI, Docker, shapefiles, GeoTIFF, Parquet
  - Developing an agent that maps natural-language analytical requests to SQL queries, database-backed retrieval, and generated outputs such as maps and tables.
  - Serves as a testbed for autonomous composition of reusable and newly synthesized software services under constrained signatures and sandboxed execution.

- **Thompson Sampling for Multi-Armed Bandits and Product Assortment** *2023*
  Tools: Python, NumPy, SciPy, Bayesian inference, conjugate priors, Monte Carlo simulation
  - Implemented and compared Thompson sampling, greedy, and UCB exploration strategies over a shared solver interface for Bernoulli and Gaussian multi-armed bandits.
  - Extended the framework to the combinatorial product assortment problem with log-normal demand and cross-product interaction effects, deriving conjugate posterior updates over the demand matrix.
  - Ran Monte Carlo simulations of action proportions and cumulative regret, documenting the exploration/exploitation trade-offs in an accompanying technical report.

Professional Experience

- **Diginavis** *Aug 2024 – Sep 2025*
  Data Scientist Innopolis, Russia
  - Built quantitative trading systems with emphasis on risk management, strategy evaluation, and performance analysis (P&L, drawdown, backtesting metrics).
  - Designed an algorithm for detecting duplicate strategy configurations, reducing redundant training/evaluation workloads by approximately 40%.
  - Worked with multi-step Dagster pipelines, S3-based storage, and highly parallel workloads across multiple compute nodes; used Python and Rust.

Technical Skills

- **Programming:** Python, Rust, SQL, Bash
- **ML / Deep Learning:** PyTorch, PyTorch Geometric, Hugging Face Transformers, discrete diffusion models (D3PM, MDLM), structured prediction, Bayesian inference and multi-armed bandits, scikit-learn, pandas, NumPy
- **LLM & RAG Systems:** LangChain, Qdrant, OpenAI API, self-hosted Qwen models, vLLM deployment on Kubernetes, prompt/tool orchestration, LLM-as-judge evaluation
- **Graph & Code Analysis:** GraphSAGE, heterogeneous graphs, AST parsing, NetworkX, UniXcoder embeddings, repository-level retrieval
- **MLOps & Infrastructure:** Docker, Docker Compose, FastAPI, Flask, Dagster, S3, Kubernetes, CI/CD, Nexus, Linux
- **Research Engineering:** benchmark design, executable evaluation, Dockerized testing environments, constrained code execution, ablation-oriented experimentation

Honors, Awards, and Presentations

- **Oral Presentation, MAS-GAIN Workshop, ASEW 2025** *2025*
  Presented GRACG: graph-augmented retrieval and Dockerized benchmark design for repository-level code generation
- **International Research Exchange, Yantai University, China** *2025*
  Invited to present ongoing research and engage with faculty and graduate students on AI/ML topics
- **Tatarstan Republic Named Scholarship** *2024*
  Awarded for academic and research achievements
- **2nd Place, Digital Breakthrough National Hackathon** *Nov 2023*
  Team-based national hackathon; built a computer-vision pipeline for vehicle classification and CO2-emission category prediction

Languages

Russian: native | **English:** IELTS 7.5; professional research presentation experience in English